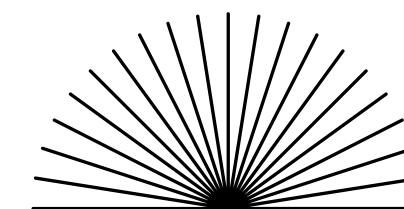


Project 6: Emerging Interpretability in LeJEPA vs. Supervised Learning

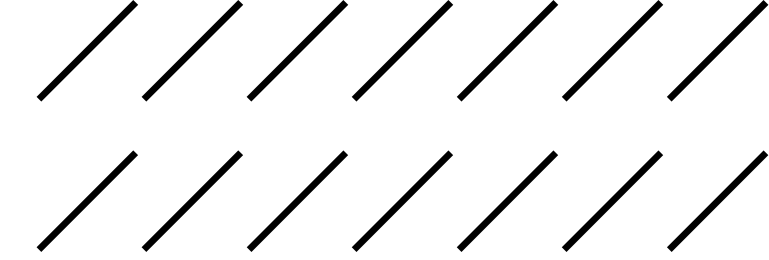
Presented By:

Noam Erez Naaman

Anastasiia Promotska

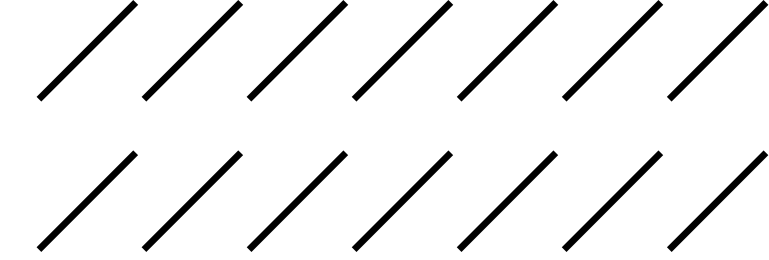


Outline



03	Problem statement
04	LeJEPA & XAI
05-07	Experimental setup
08	Methodology Pipeline
09-11	Evaluation
12-21	Results
22	Conclusions
23	References

Problem statement



Task

A comparative layer-by-layer analysis of feature localization and semantic consistency between LeJEPA trained models and End-to-End Supervised models, utilizing both Vision Transformers (ViT) and Convolutional Neural Networks (CNNs).

”

Do LeJEPA-trained models produce layer-wise latent features whose PCA-derived semantic regions align better with XAI saliency maps than supervised models?

”

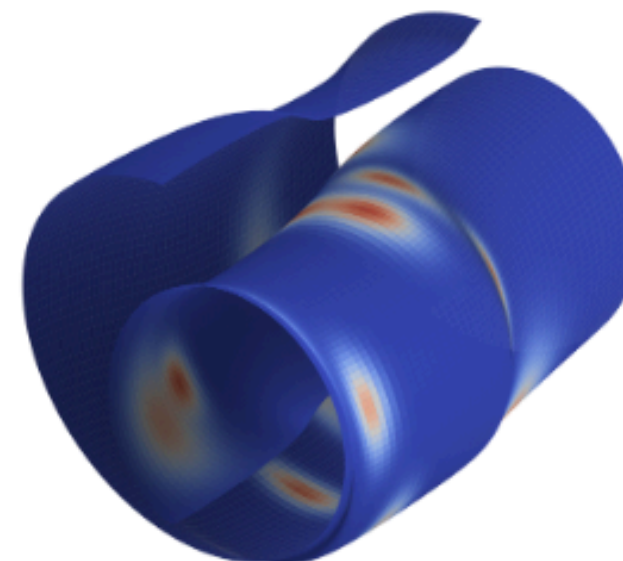
LeJEPA & XAI

JEPA learns semantic meaning by predicting in embedding space

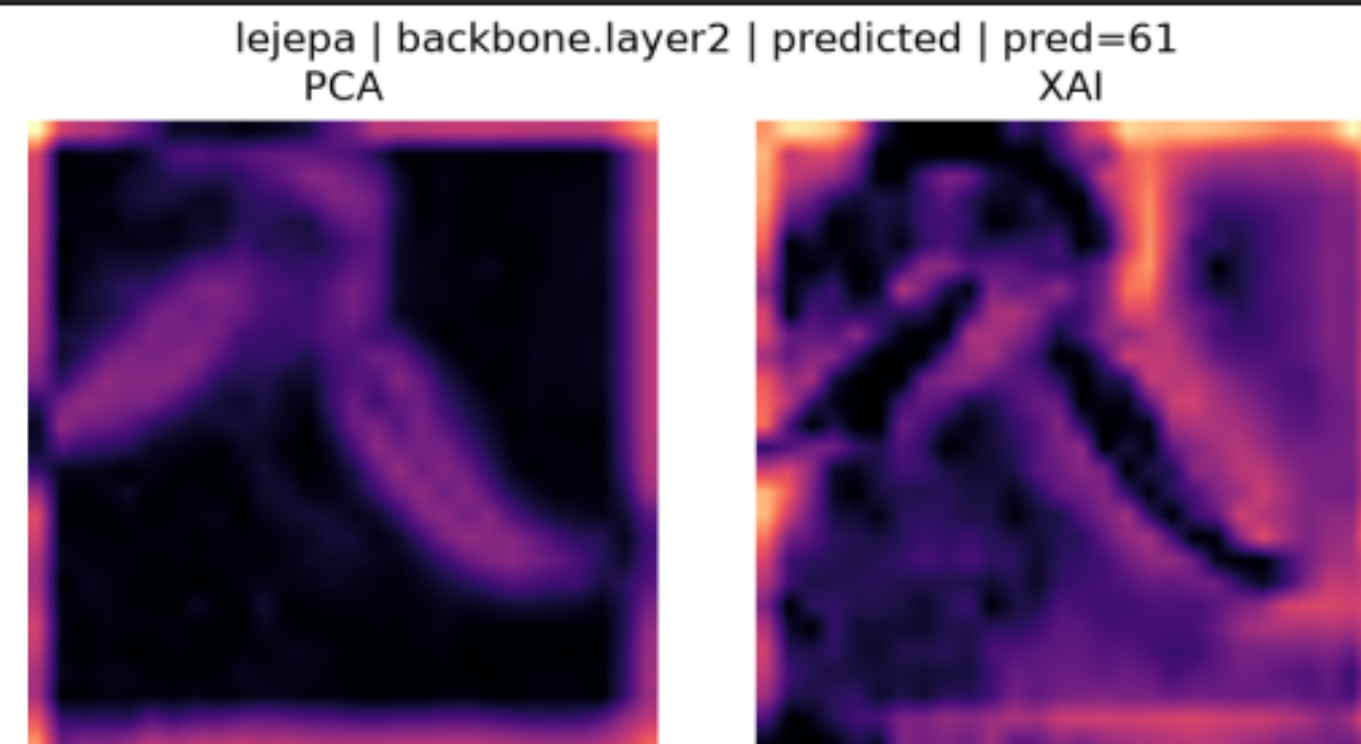
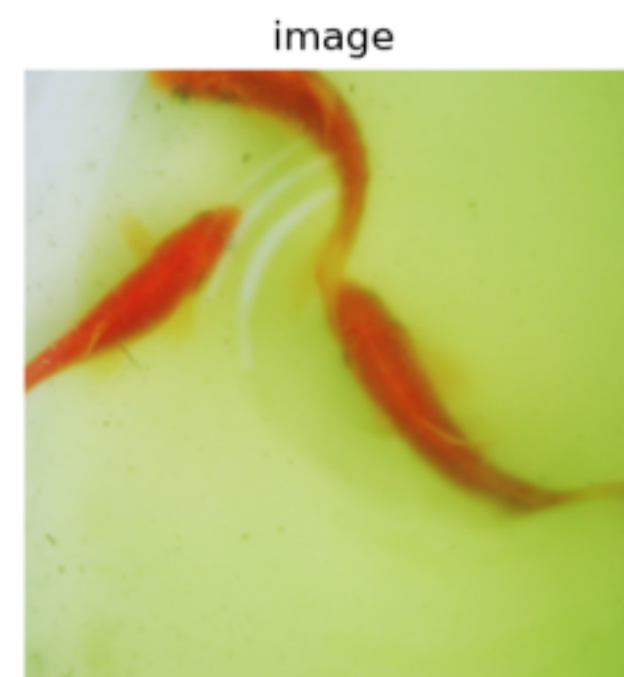
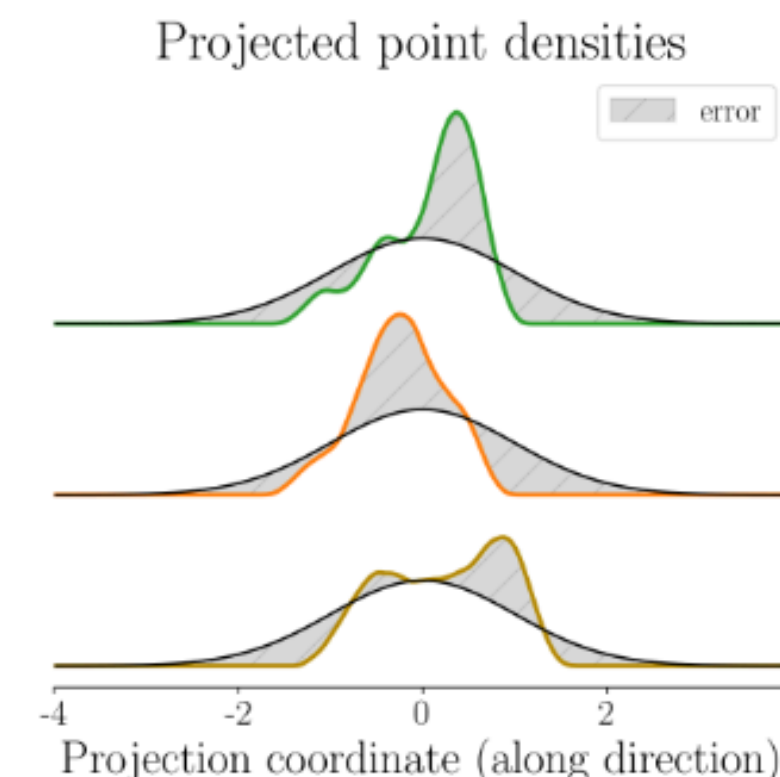
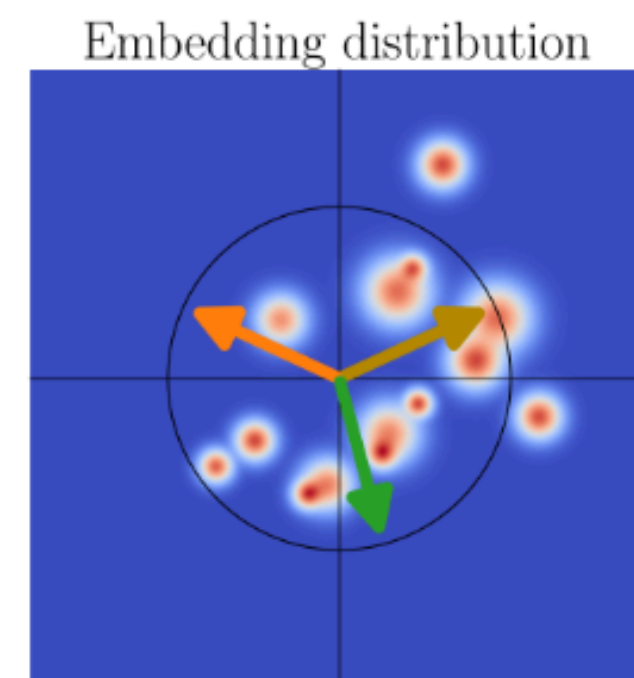
LeJEPA improves stability with SIGreg, driving distribution to isotropic Gaussian

Semantic maps show unsupervised emergent semantic segmentation

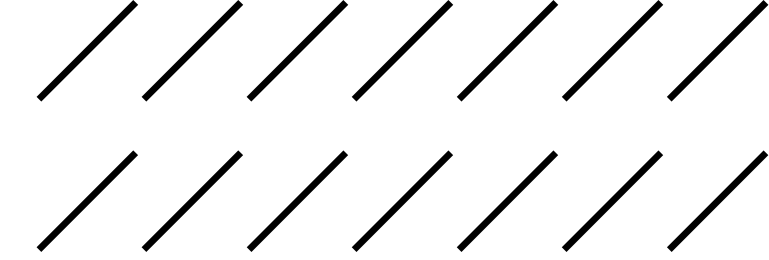
XAI saliency maps show where the model is “looking” to make classification decision



f_{θ}
→



Experimental setup - Datasets selection



CIFAR10

small-resolution setting
10 classes
32 x 32 RGB images

baseline

ImageNet-100

Subset of ImageNet-1K
100 classes
224×224 RGB images

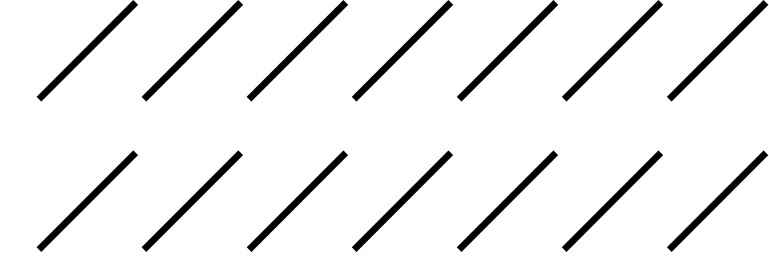
**high-resolution
benchmark**

ImageNet-S-50

50 classes
also provides segmentation
masks

grounded evaluation

Experimental setup - Models selection



Controlled experiments

ResNet18

8 residual blocks
4 stages
512 final feature dimension

preserves spatial structure
useful for layer-wise
analysis

**standard convolutional
baseline**

ViT-Tiny

patch size 4
embedding dimension 192
6 transformer blocks
3 attention heads

For CIFAR-10

ViT-S/16

patch size 16
embedding dimension 384
12 transformer blocks
6 heads

**For ImageNet-100 and
ImageNet-S-50**

External reference

ViT-B/16

85.8 million backbone
parameters

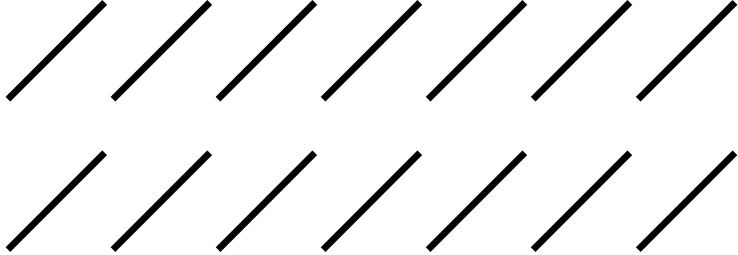
**public available
Supervised + LeJEP
checkpoints**

“

Does the semantic-alignment
behavior observed in our
smaller controlled models also
appear in larger ImageNet-1K
pretrained models?

”

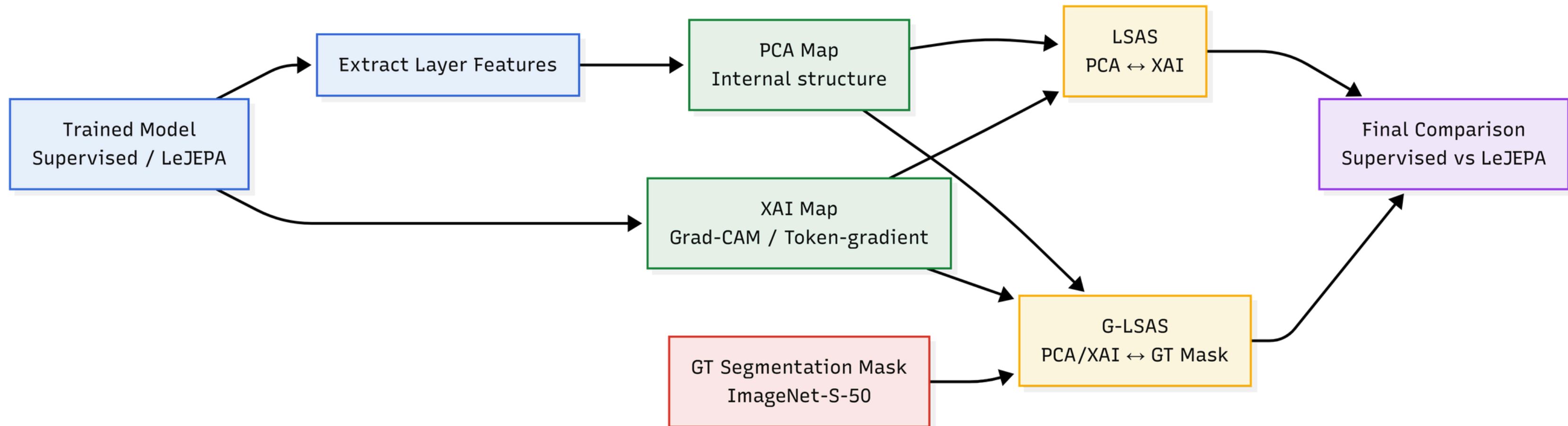
Experimental setup - Comparative fairness



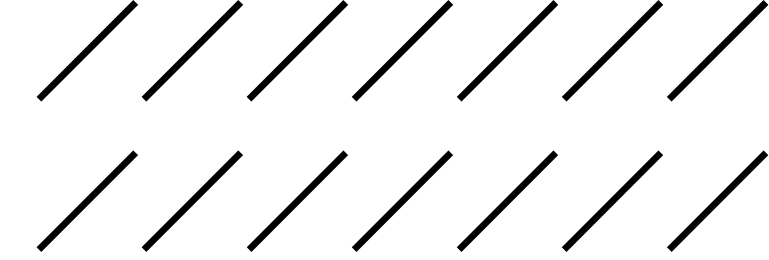
Dataset	Model/architecture	Method	Experiment	Backbone	Head	Eval total	Training total	Classes	Input
CIFAR-10	ResNet-18, CIFAR stem	supervised	c10-r18-sup	11.17M	0.01M	11.17M	11.17M	10	32
CIFAR-10	ResNet-18, CIFAR stem	LeJEPA	c10-r18-lejepa-v5-sigreg-002	11.17M	0.01M	11.17M	11.83M	10	32
CIFAR-10	ViT-Tiny / patch4	supervised	c10-vit-sup	2.69M	0.00M	2.69M	2.69M	10	32
CIFAR-10	ViT-Tiny / patch4	LeJEPA	c10-vit-lejepa-v4-sig005	2.69M	0.00M	2.69M	3.19M	10	32
ImageNet-100	ResNet-18, ImageNet stem	supervised	in100-r18-sup	11.18M	0.05M	11.23M	11.23M	100	224
ImageNet-100	ResNet-18, ImageNet stem	LeJEPA	in100-r18-lejepa	11.18M	0.05M	11.23M	11.84M	100	224
ImageNet-100	ViT-S/16	supervised	in100-vit-sup	21.67M	0.04M	21.70M	21.70M	100	224
ImageNet-100	ViT-S/16	LeJEPA	in100-vit-lejepa	21.67M	0.04M	21.70M	22.26M	100	224
ImageNet-S-50	ResNet-18, ImageNet stem	supervised	ins50-r18-sup	11.18M	0.03M	11.20M	11.20M	50	224
ImageNet-S-50	ResNet-18, ImageNet stem	LeJEPA	ins50-r18-lejepa	11.18M	0.03M	11.20M	11.84M	50	224
ImageNet-S-50	ViT-S/16	supervised	ins50-vit-sup	21.67M	0.02M	21.68M	21.68M	50	224
ImageNet-S-50	ViT-S/16	LeJEPA	ins50-vit-lejepa	21.67M	0.02M	21.68M	22.26M	50	224
ImageNet-100	Public LeJEPA ViT-B/16	LeJEPA	public-okai-vitb16	85.80M	0.00M	85.80M	–	0	224
ImageNet-100	Public supervised ViT-B/16	supervised	public-supervised-vitb16	85.80M	0.77M	86.57M	–	1000	224

Methodology Pipeline

04/10



Evaluation - Representation quality



Classification accuracy

For **supervised** models.

Standard validation accuracy of the classifier trained end-to-end with class labels

Linear probe accuracy

For **LeJEP**A models

Linear classifier, trained on top of the frozen features

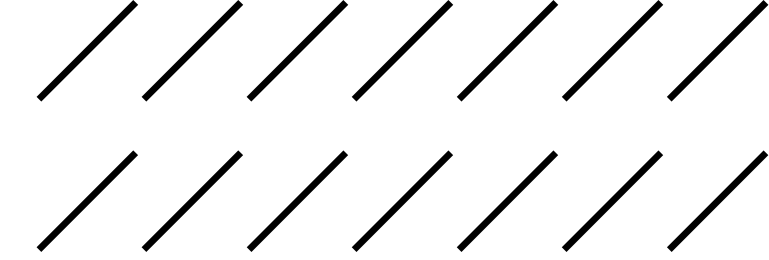
Tests whether the learned representation contains class-discriminative information

KNN accuracy

For each extracted frozen validation feature:

Find the closest training features and predict the class by majority vote

Evaluation - Interpretability and semantic alignment



LSAS - Layer-wise Semantic Alignment Score

$$LSAS_l = 0.5 \cdot \text{Corr}(PCA_l, XAI_l) + 0.5 \cdot \text{SoftIoU}(PCA_l, XAI_l)$$

Measures how well a model's internal feature structure aligns with its explanation map at a given layer.

For each selected layer, there are computed:

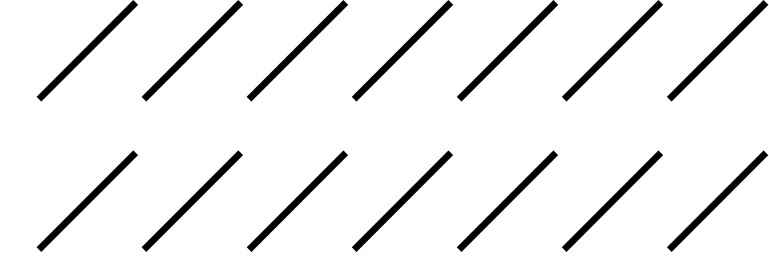
- PCA map from layer features
- XAI map from model explanation

Where:

- Corr = whether PCA and XAI maps vary together spatially
- SoftIoU = how much the two heatmaps overlap

higher → better

Evaluation - Interpretability and semantic alignment



G-LSAS – Grounded Layer-wise Semantic Alignment Score

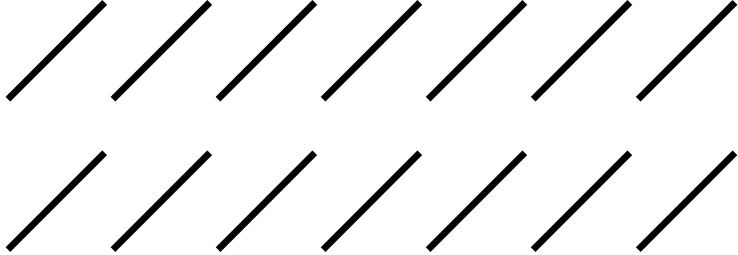
$$\text{G-LSAS}_l = 0.35 \cdot \text{Corr}(\text{PCA}_l, \text{XAI}_l) + 0.25 \cdot \text{IoU}(\text{PCA}_l, \text{GT}) + 0.25 \cdot \text{IoU}(\text{XAI}_l, \text{GT}) + 0.15 \cdot \text{Stability}(\text{PCA}_l, \text{PCA}_l^{\text{aug}})$$

Extends LSAS by adding ground-truth segmentation masks

Checks whether the PCA feature map and XAI map agree with each other and whether both align with the real object region

The score includes PCA-XAI correlation, PCA-to-mask overlap, XAI-to-mask overlap, and PCA stability under augmentation

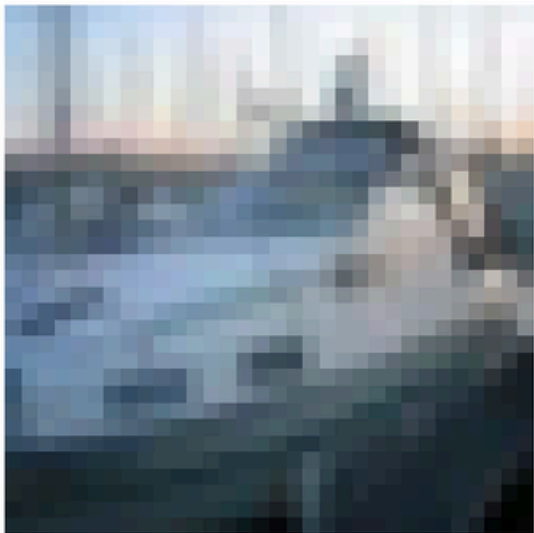
Qualitative registers comparison



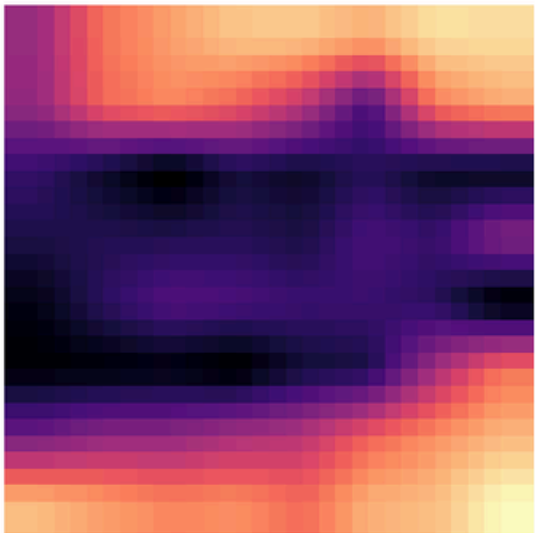
Without registers

vit_lejepa | token_gradient | blocks.4 | target=predicted | true=8 pred=8 cam=8

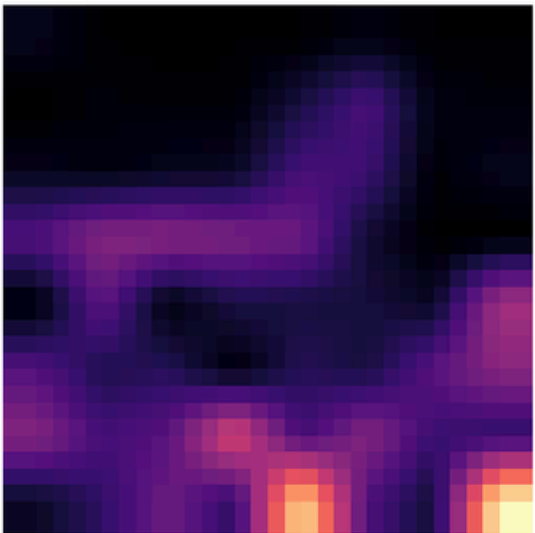
image



PCA mask



Rollout



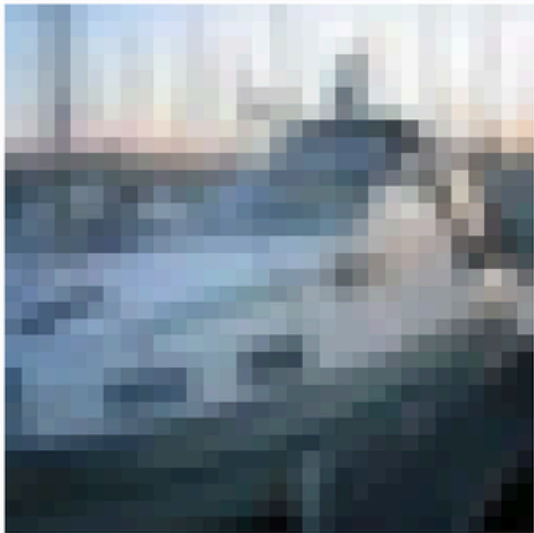
overlay



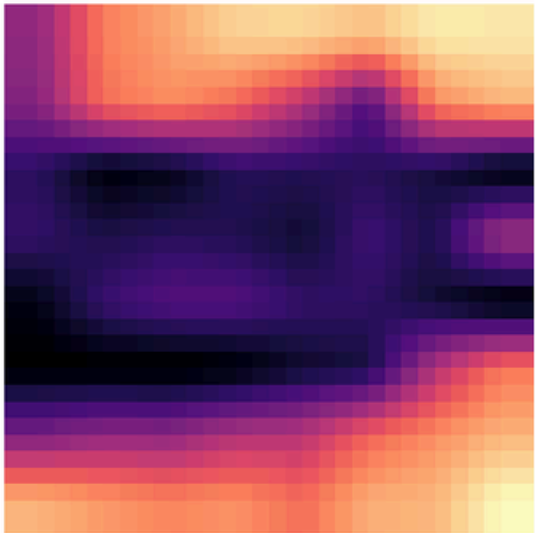
With registers

vit_lejepa | token_gradient | blocks.4 | target=predicted | true=8 pred=8 cam=8

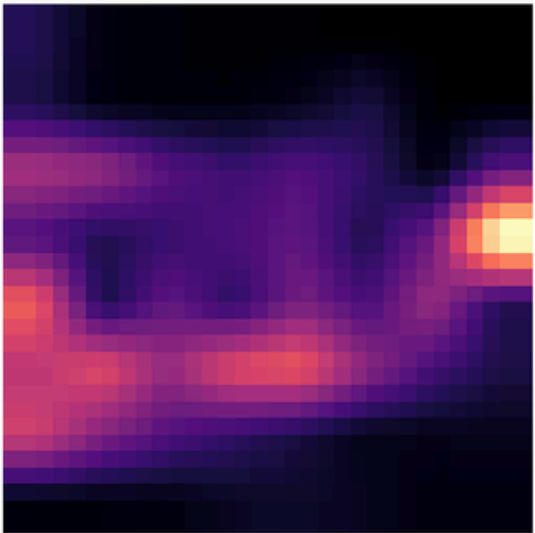
image



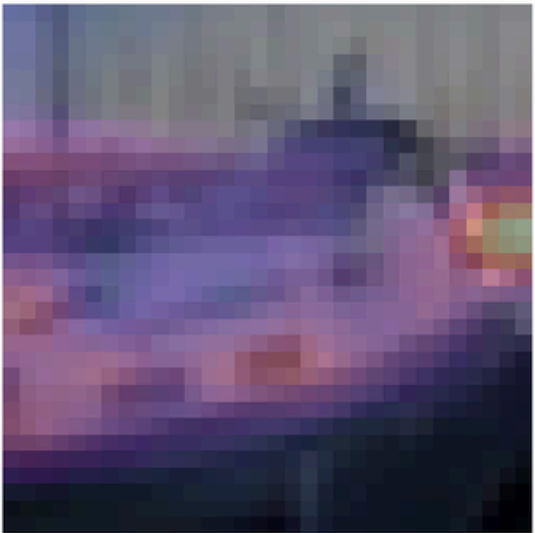
PCA mask



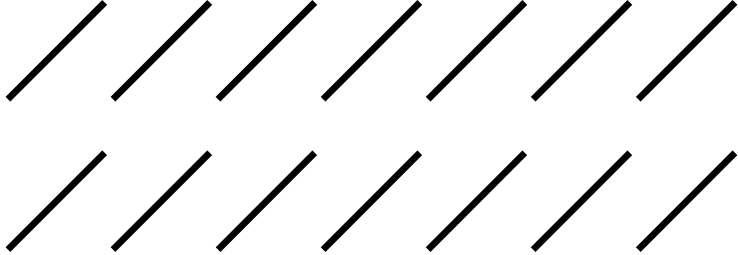
Rollout



overlay



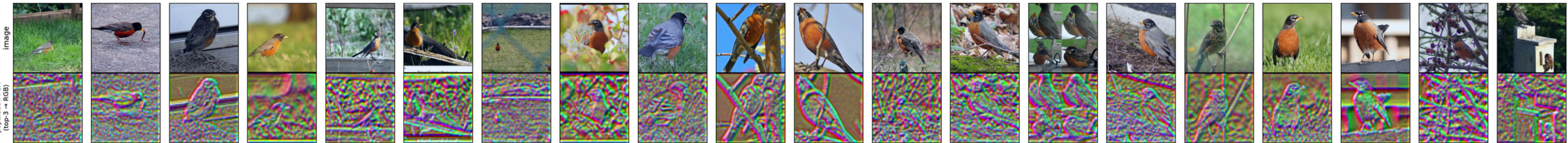
Qualitative PCA segmentation



Imagenet-100, LeJEPA resnet

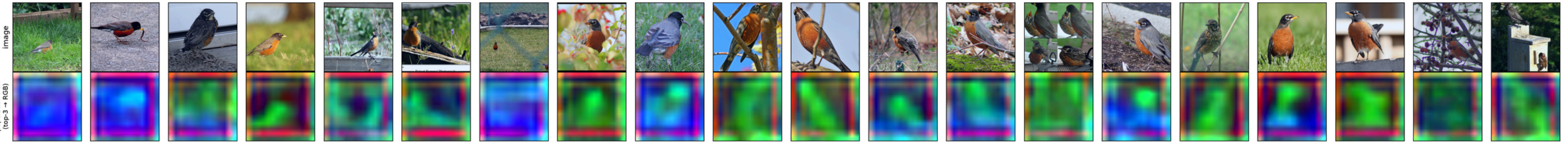
Layer 1

lejepa · backbone.layer1 · shared PCA basis (same color = same direction across images)



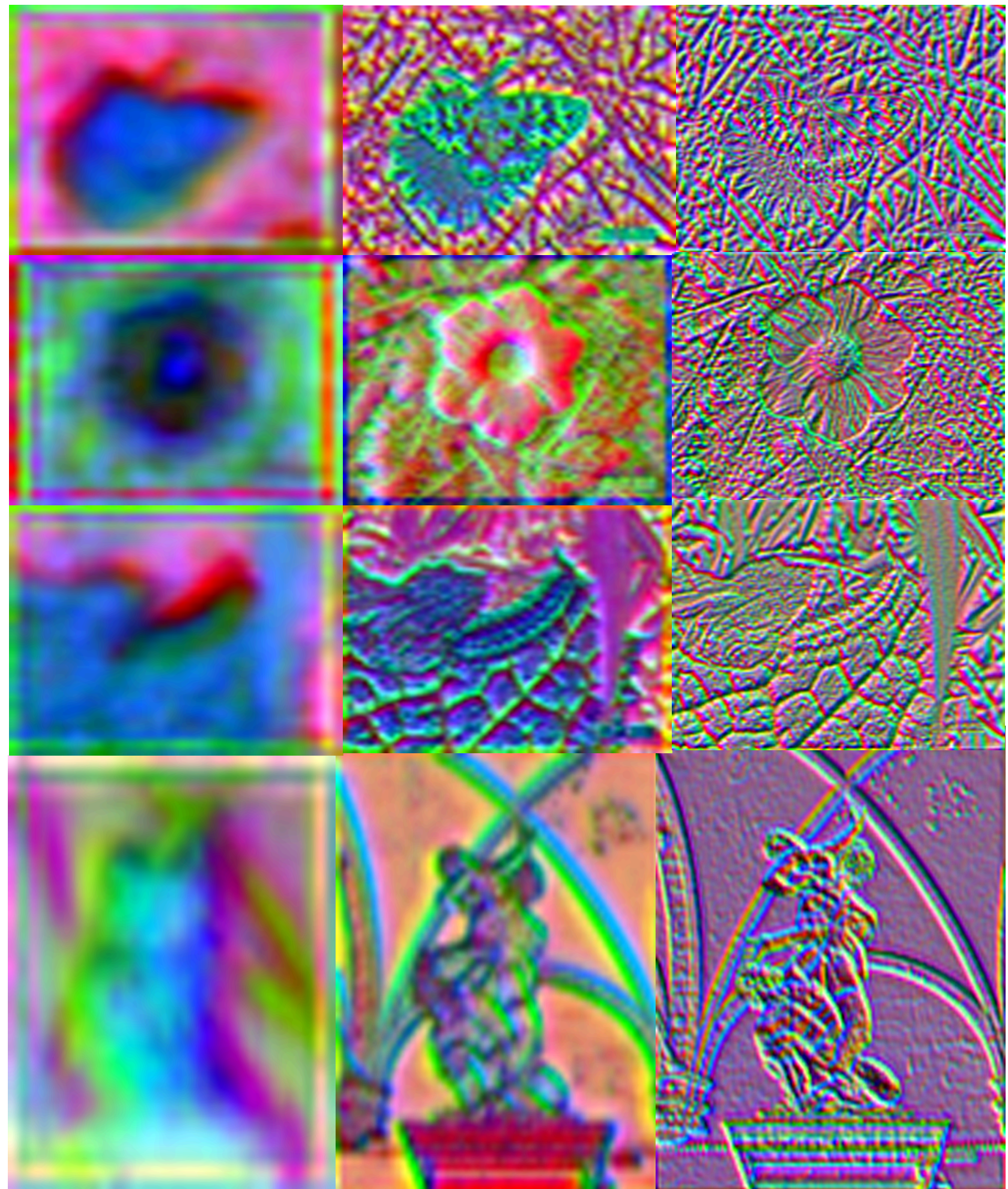
Layer 3

lejepa · backbone.layer3 · shared PCA basis (same color = same direction across images)

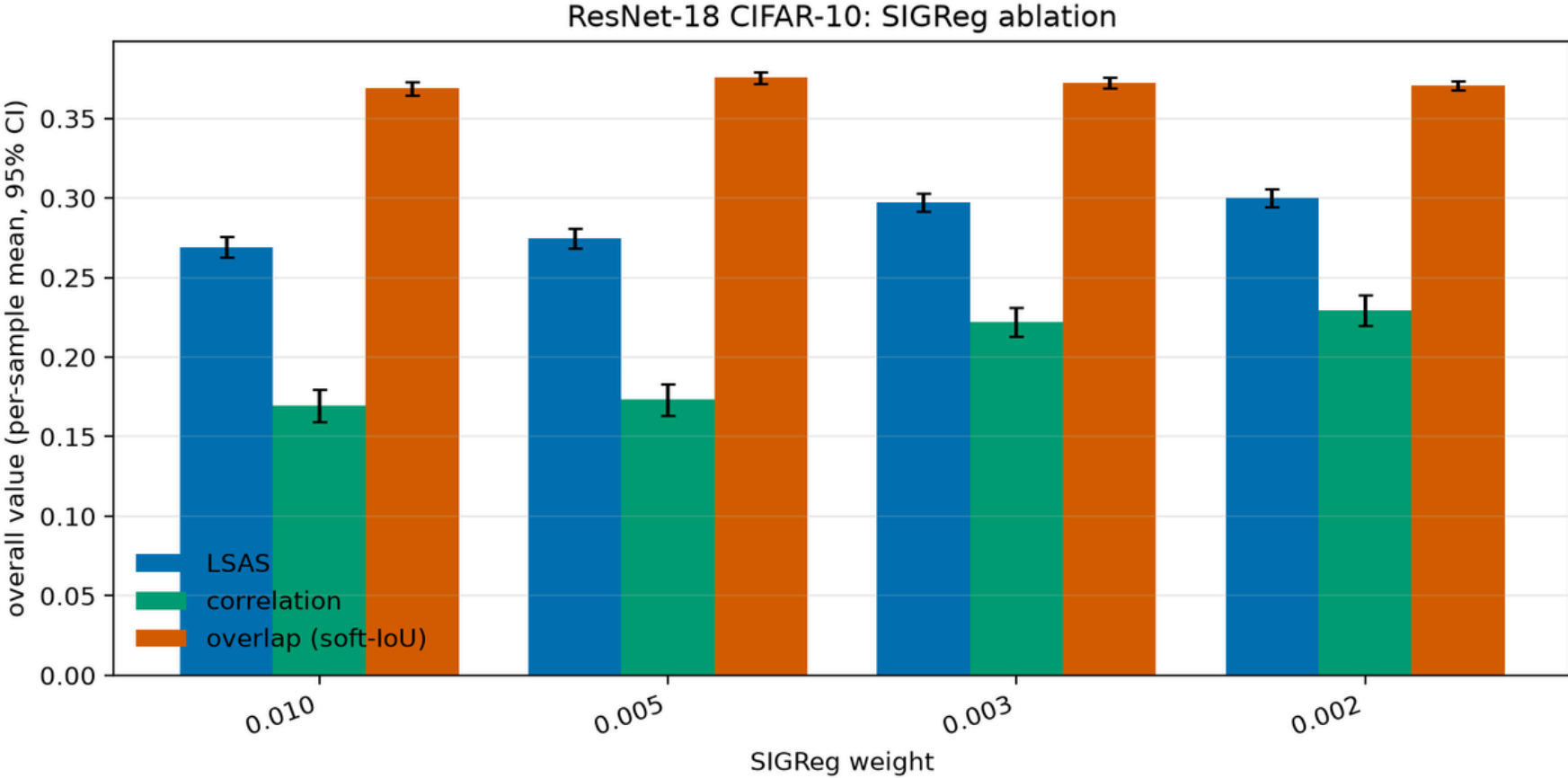
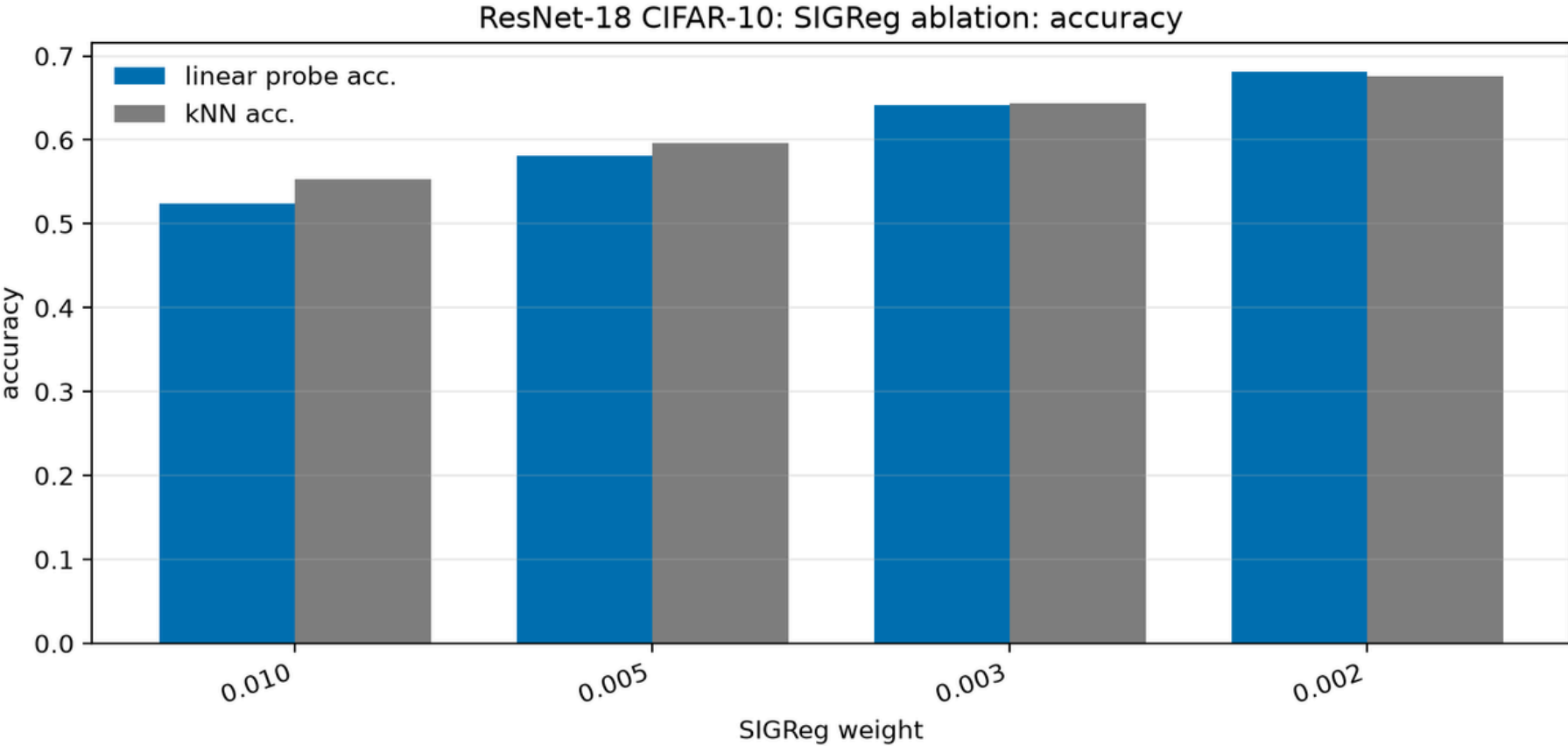
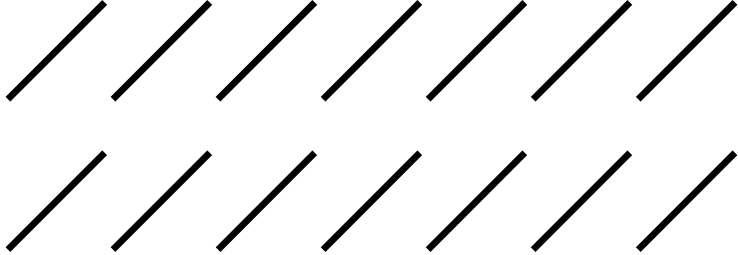


Qualitative PCA Layer Comparison

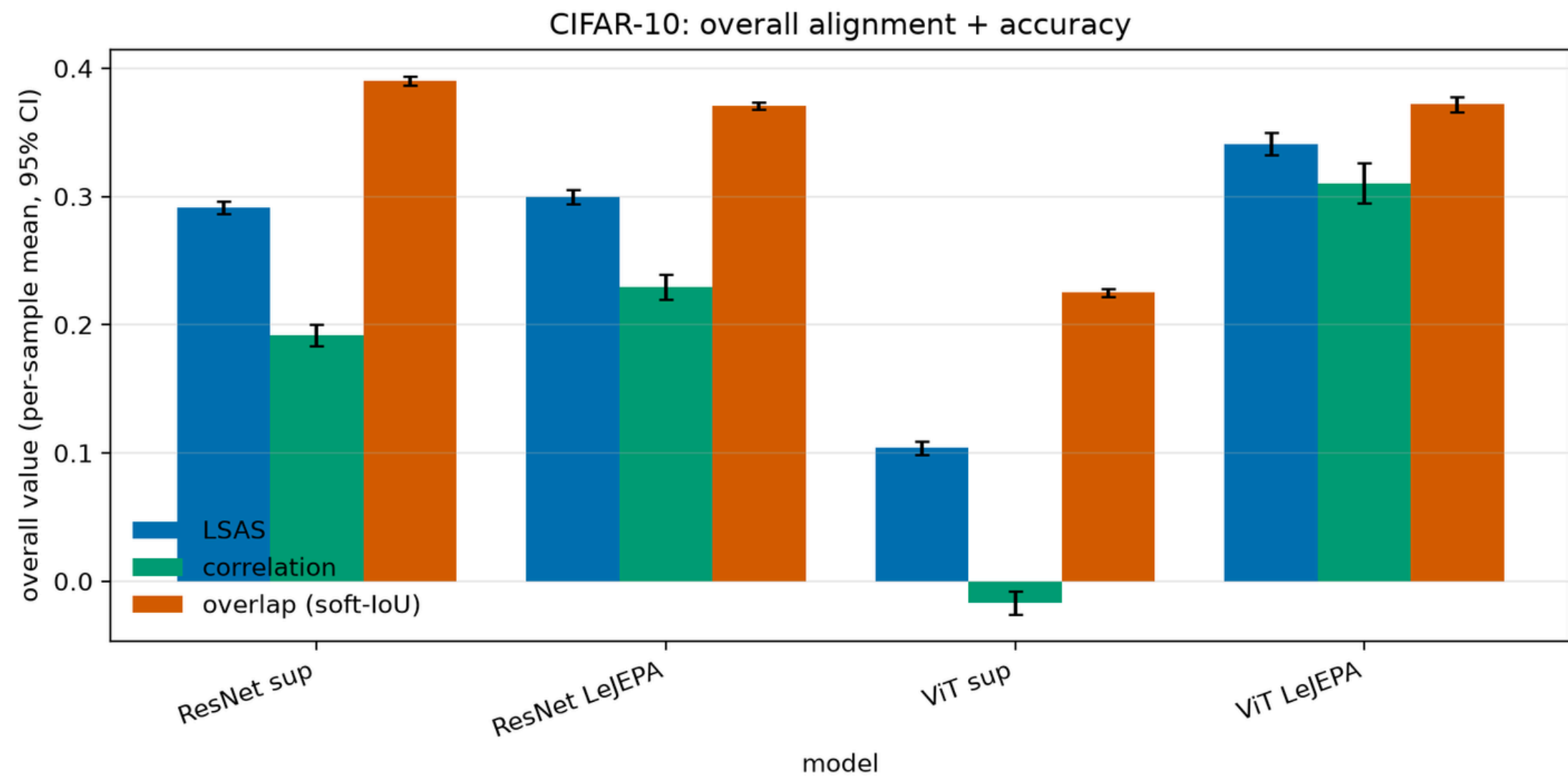
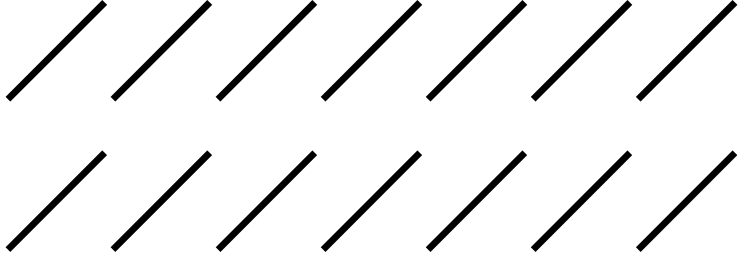
New unseen photos (taken by
us), LeJEPA resnet



SIGreg ablation



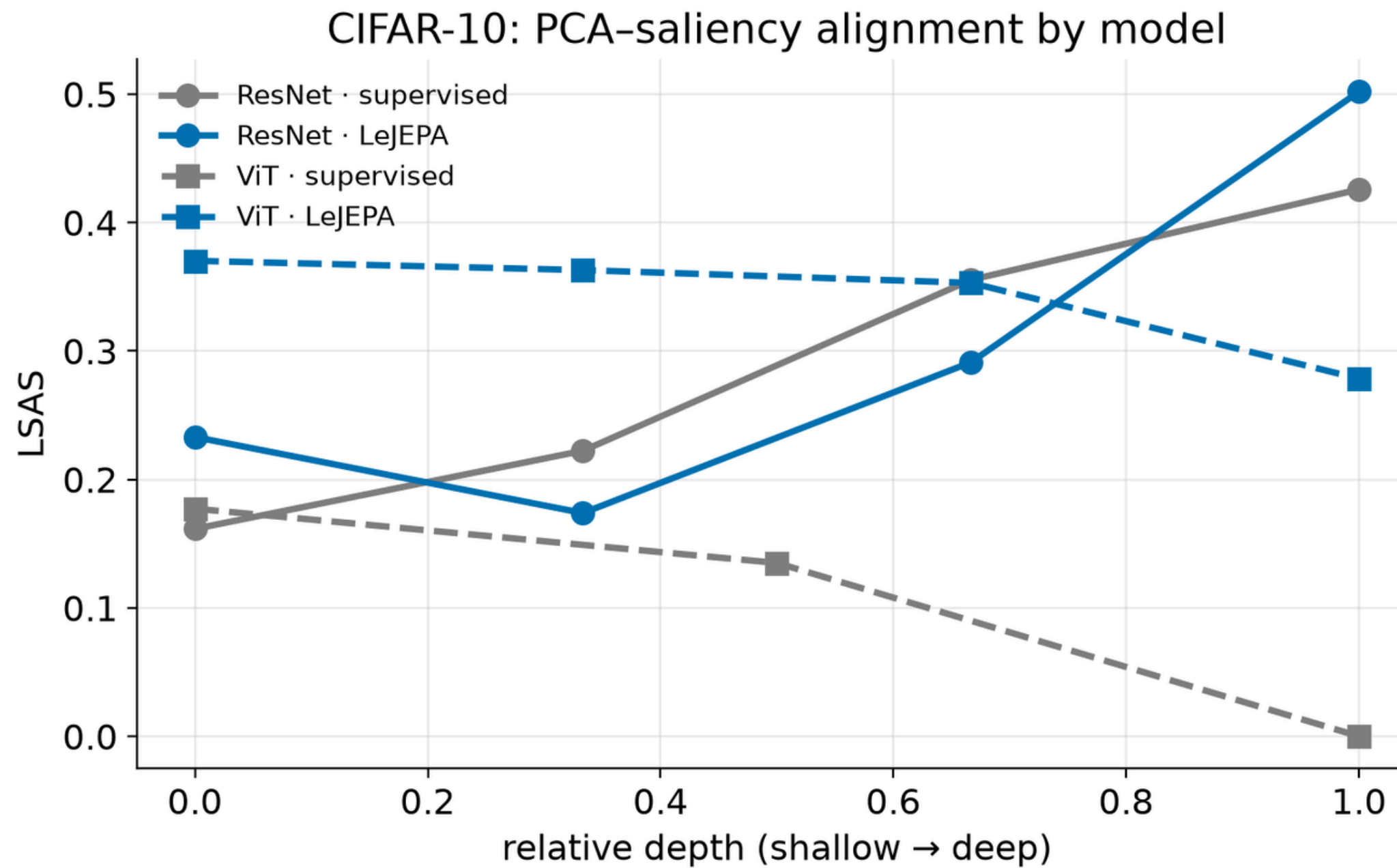
Results - CIFAR-10 overall



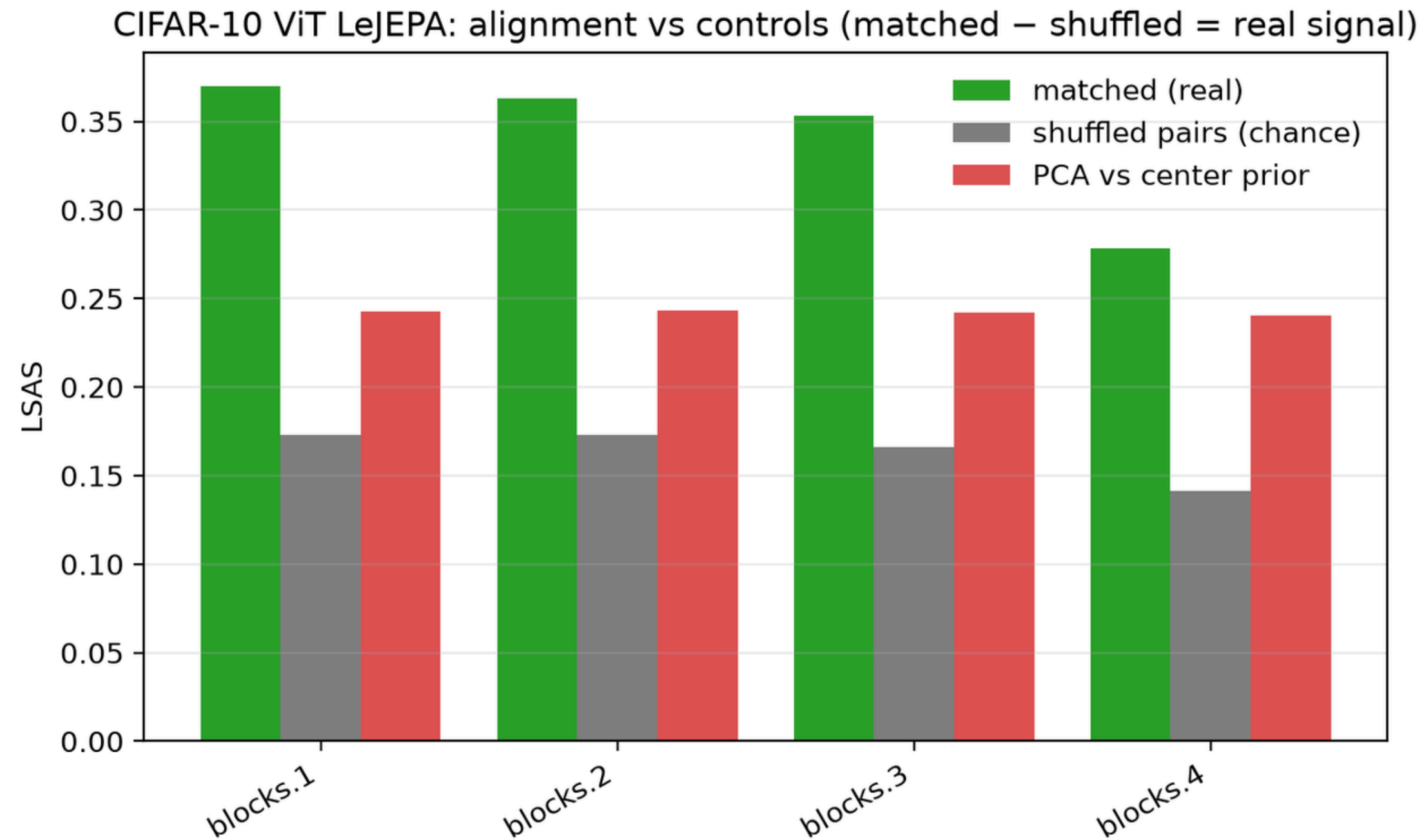
CIFAR-10 — model summary

model	probe acc	kNN acc	mean lsas	peak lsas
ResNet supervised	0.912	0.913	0.291	0.426
ResNet LeJEPA	0.681	0.676	0.300	0.502
ViT supervised	0.804	0.812	0.104	0.177
ViT LeJEPA	0.390	0.427	0.341	0.370

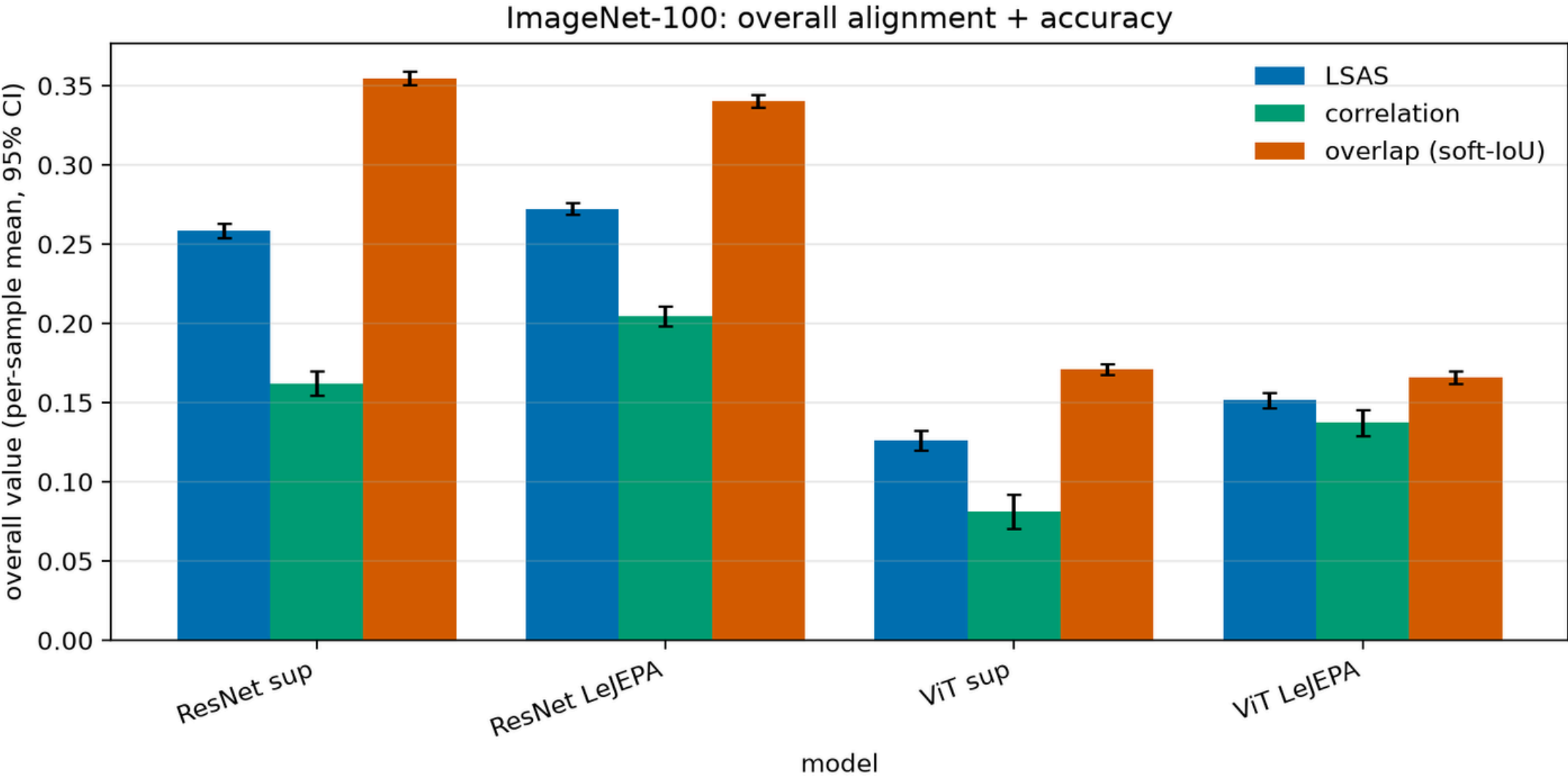
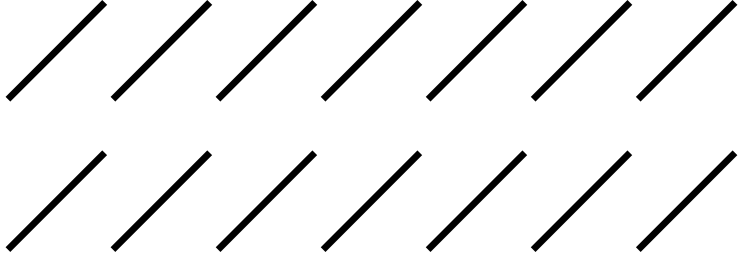
Results - CIFAR-10 by layer



Results - CIFAR-10 shuffle/center-bias



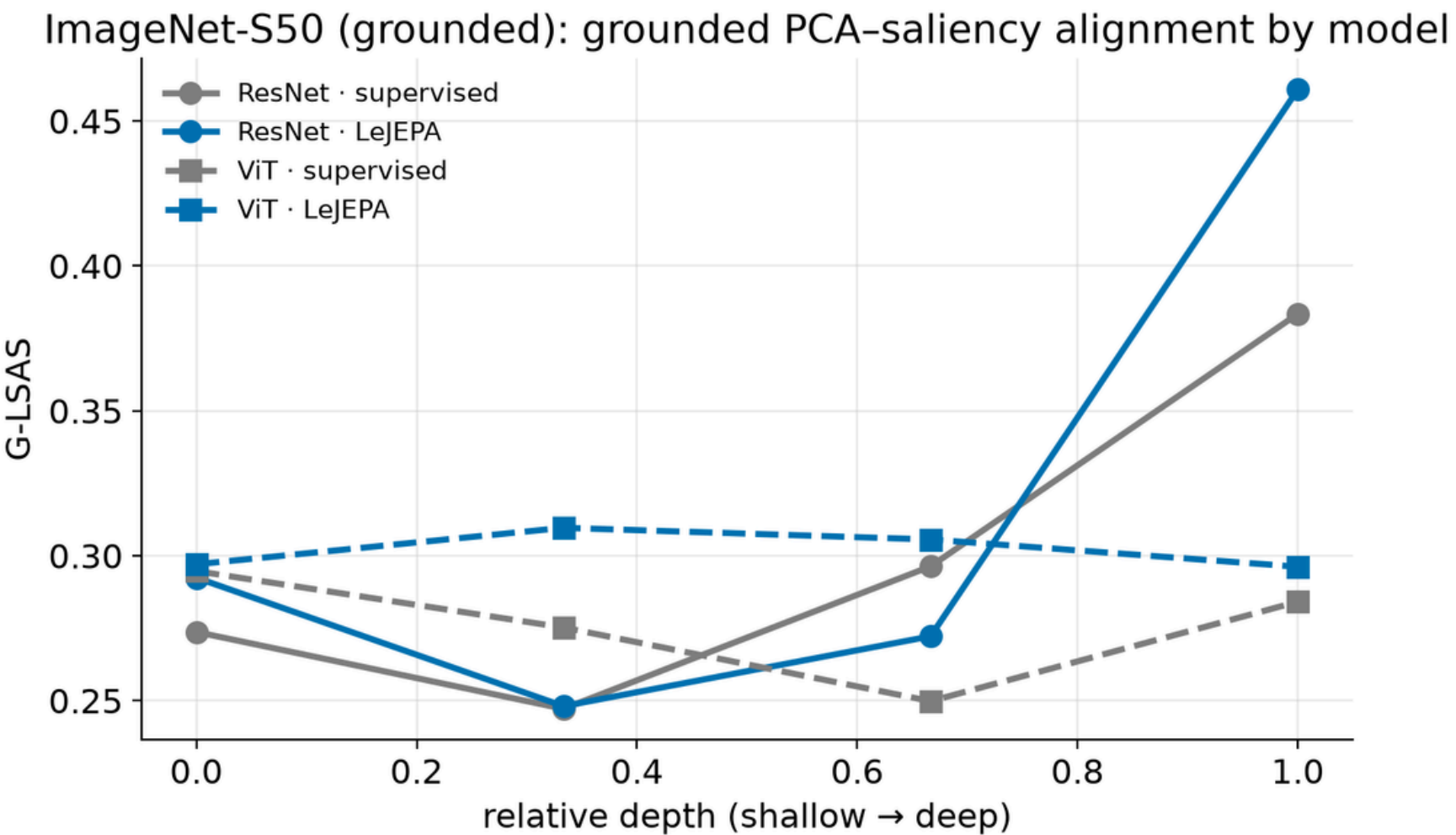
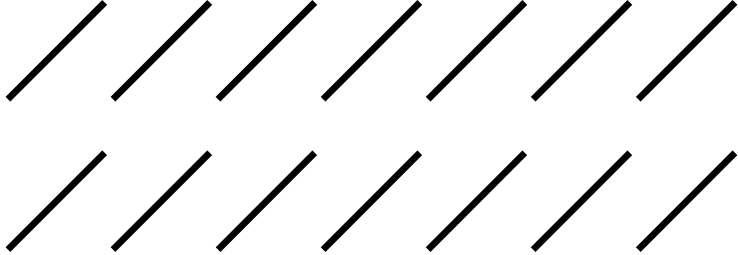
Results - Imagenet-100 overall



ImageNet-100 — model summary

model	probe acc	kNN acc	mean lsas	peak lsas
ResNet supervised	0.794	0.795	0.258	0.447
ResNet LeJEPA	0.317	0.308	0.272	0.581
ViT supervised	0.573	0.579	0.126	0.193
ViT LeJEPA	0.253	0.201	0.152	0.214

Results - Imagenet-S50 grounded analysis



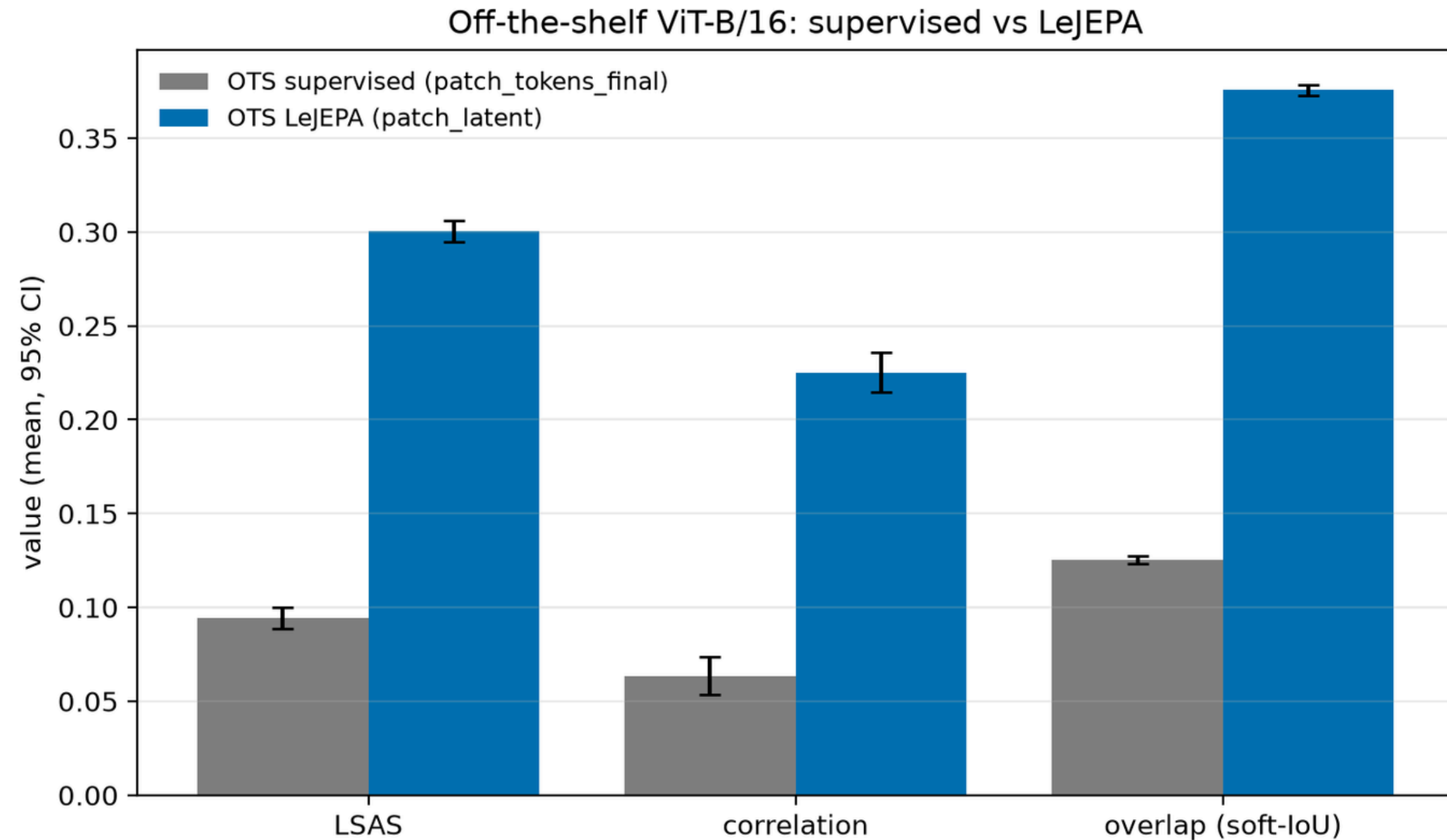
ImageNet-S50 (grounded) — model summary

model	probe acc	kNN acc	mean g_lsas	peak g_lsas	IoU(PCA,mask)	IoU(saliency,mask)
ResNet supervised	0.820	0.824	0.300	0.383	0.208	0.238
ResNet LeJEPA	0.513	0.495	0.318	0.461	0.183	0.243
ViT supervised	0.622	0.637	0.276	0.295	0.195	0.252
ViT LeJEPA	0.374	0.327	0.302	0.310	0.200	0.248

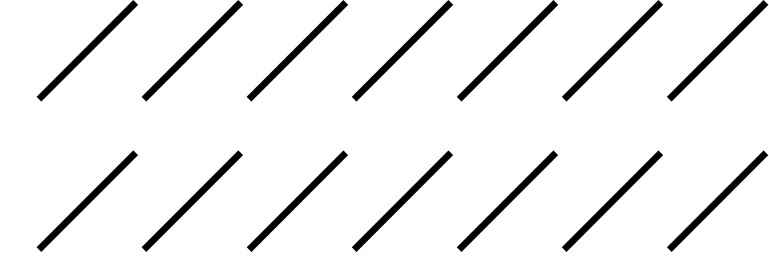
Results - Public JEPA vs Supervised

ViT-B/16

85.8 million backbone parameters



Conclusions



Our results:

- We see better interpretability scores overall for JEPA over supervised
- Grounding with real segmentation shows good results

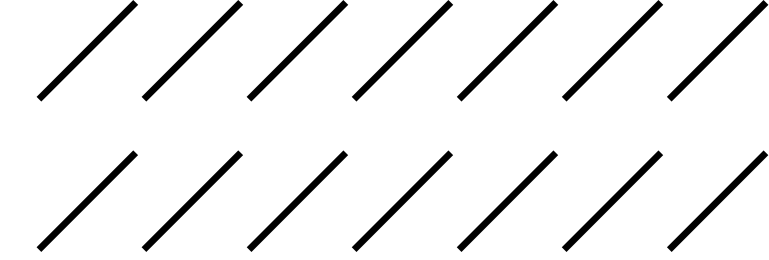
Issues:

- Layer by layer scores vary
- Trade-off between IoU and correlation, SAS and accuracy
- Public models outperform ours but are much bigger

Future improvements:

- Run JEPA with multiple seeds
- Larger scope, more complex datasets
- More grounding measures

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- Darcet, T., Oquab, M., Mairal, J., & Bojanowski, P. (2024). Vision Transformers Need Registers. ICLR. arXiv:2309.16588.
- Darcet, T., Oquab, M., Mairal, J., & Bojanowski, P. (2024). Vision Transformers Need Registers. ICLR. arXiv:2309.16588. (your register experiment)
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